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Conference Paper · February 2023

DOI: 10.1109/IMCERT57083.2023.10075119

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Assessment of Advanced Artificial Intelligence Techniques for Flood Forecasting

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2023 International Multi-disciplinary Conference in Emerging Research Trends (IMCERT) | 979-8-3503-4770-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/IMCERT57083.2023.10075119

***Abstract*— Flooding is a natural calamity that can destroy people's lives, infrastructure, and the economy. Forecasting floods is critical for providing people with long-term flood risk management. Flood forecasting is essential in providing early information and knowledge to decision-makers to reduce the impact of flooding. The warning can also be given to potential flood victims and locations, and necessary action, such as mitigation and evacuation, can be taken. With current estimates showing increasing future scenarios, comprehensive flood risk management measures, including flood modelling, are needed. This publication aims to analyses flood risks worldwide. Various AI techniques have been developed and deployed to predict floods and take preventative actions. The primary goal of this study is to assess current improvements in flood forecasting utilizing artificial neural networks (ANNs), adaptive neuro-fuzzy inference systems (ANFIS), support vector machines (SVMs), and k-nearest neighbors (KNNs). As a result, this research presents the most effective short and long-term flood modelling techniques. ANNs, ANFIS, and SVMs are the most successful solutions for forecasting floods. Finally, new research and development directions are suggested to predict floods and take preventative actions.**

Keywords— Climate Change, Weather, Artificial Intelligence, Modeling

I. INTRODUCTION

Due to rapid population growth and global warming, urban flooding has become an increasingly important hazard risk for humans in recent decades. Floods occur annually throughout the world due to their geographical layout, population density, and unequal economic and social development, particularly in Asia and the Pacific. [1]. During the last two decades, from 2000 to 2019, there were an average of 367 global disasters, primarily floods and storms (44% and 28%, respectively). Asia experienced the most significant number of natural disasters. From 2000 to 2019, Asia faced 57.24 % of all disasters. The frequency and severity of natural calamities in Asia are mainly attributable to the continent's vastness and landforms that provide a high risk of natural hazards, including major rivers, flood plains, and earthquake fault lines [1, 2]. At this time, the provision of an objective model for predicting river flow can assist in the decision of relevant tasks such as (i) the appropriate design of water drains and storage networks, (ii) the observing of extreme events such as droughts and floods, (iii) the strategy of future reservoir extension or contraction, (iv) aiding in the avoidance and knowledge of hydrologic dangers such as the modification of the hydroclimatic regime, (v) erosion and silt movement, and sediment [3]. Despite countless tools, methods, and ideas, we cannot afford utilities large data mine effectively.

Because of the human setup and management of the data, the current forecasts of how the earth would adapt to global change are riddled with subjective and unrealistic assumptions. That's because the configuration and management of the data are done manually. There is already a great deal of rainfall-runoff models spanning various categories that can be used for flow forecasting. Most rainfall-runoff systems have been designed and built with the process's physical or systems-theoretical parts in mind [4]. The current hydrological analysis is complicated by issues in water levels forecasting analysis, such as topographic data and the optimization of model parameters for the region. Recent developments in Artificial Intelligence (AI) technology have led to the implementation AI technology in water resource studies [5]. Numerous degrees of success have resulted from developing various AI models originating from multiple domains of knowledge, employing a variety of methodologies, and pursuing distinct objectives. Several modelling methods and black-box alternatives are available in hydrology, even though many are variants of ANN architectures and neuro-fuzzy techniques. According to studies, the ANN model also performs well in flood forecasting in tropical and subtropical regions [6]. The primary objectives of this paper are to (1) assessment of AI techniques and (2) evaluation of AI techniques for flood forecasting in tropical regions. This paper comprises the following sections: 1) background and objectives, (2) current progress in flood forecasting, (3) overview of ai techniques, (4) selected advanced ai techniques, (5) discussions of past employed ai for flood forecasting, (6) warning and dissemination, and (7) future directions/conclusion.

II. CURRENT PROGRESS IN FLOOD FORECASTING

In recent decades, numerous studies on different theories and concepts have been conducted to build flood forecasting systems effectively. Because many natural phenomena can cause floods in tropical regions, it is crucial to analyze the conditions unique to that area to develop a solution that considers that region's dynamics. Several dynamic factors influence water flow; successful flood severity forecasting depends more on data-driven modelling. Several raster models based on physical phenomena were used to forecast hydrological occurrences, and statistical models were used. The study's most recent updates assert that AI models perform better than conventional statistical models and show the effectiveness of precisely predicting the necessary parameters for flood forecasting. Statistical modelling has been a tire so machine learning (ML) and deep learning (DL) models that can handle complex data and develop correlations between multiple physical aspects. As a result, ML and DL models are becoming increasingly popular. It is not necessary to be conversant with the underlying physical properties and their mathematical correlations to use ML techniques for flood forecasting. These techniques can be found here. It is possible to train a hybrid model utilizing signals data and yet have it produce promising findings far in advance of a flood event [7]. Many techniques were used in the past, as mentioned in fig 1.

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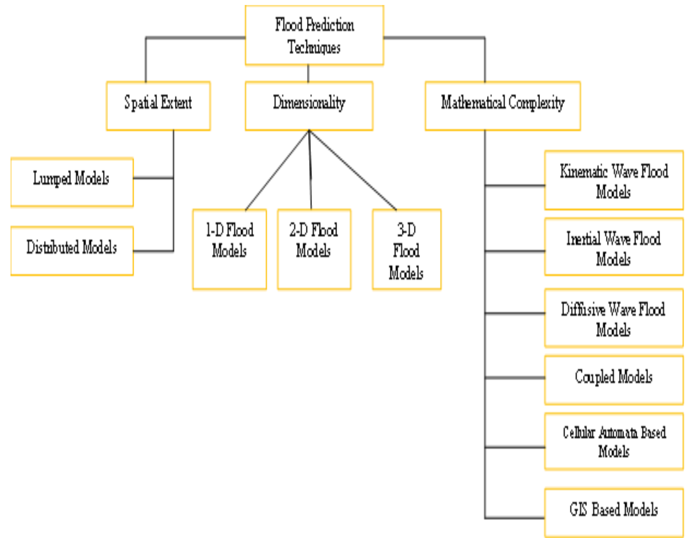

Fig 1 Flood modelling techniques [8]

III. OVERVIEW OF ARTIFICIAL INTELLIGENCE (AI)

This section presents an overview of AI techniques that have been utilized in hydrology in the past and that are relevant to the issue of river flow forecasting. For scientists, the phrase artificial intelligence (AI), which is a phrase used in so many diverse contexts, can be confounding. The applications of AI techniques are much more extensive and widespread than the more well-known ones, such as ML and DL. Broadly, ML is considered a subset of AI, while DL is a subset of ML, as illustrated in fig 2. These include conventional, and computational-based (ANNs, ANFIS, SVM, fuzzy logic, and evolutionary calculation) approaches [9]. This section will briefly introduce advanced AI techniques, including ML and DL, commonly employed in hydrology, specifically for flood prediction. Covering the entire AI universe would be impractical, so that this section will focus on flood forecasting.

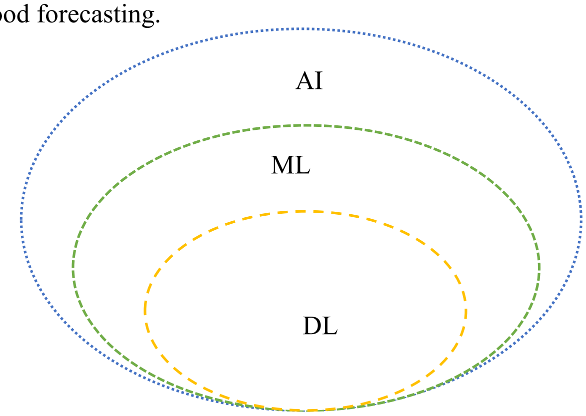

Fig 2 Basic Classification of AI (Machine Learning is a type of Artificial Intelligence. Deep Learning is an exceptionally complex part of Machine Learning)

A. Machine Learning (ML)

The flow of building an ML model is shown in fig 3. This flow represents a general flow, with variations across researchers involving the usage of observational datasets without data analysis and a validation phase between training and testing. Credible sources acquire data for creating an ML model for long-term flood predictions. After that, the first datasets are reformatted into a different format and made ready for input into the model. Examining the data is extremely important since it decides which information to employ in the model. During the creation stage of the model, many tools and programming languages were utilized to translate an algorithm into an operational model. After that, this is willing to be scrutinized and undergo training and testing [10]. The development of a single or hybrid model using ML for short-term and long-term flood predictions can be considered two distinct approaches. When making predictions, a single model will only employ one type of machine learning, but a hybrid model will utilise numerous ML methods in the integration or ensembles. Integration consisted of integrating one machine learning model with another ML model and an int learning model with a data-driven model, a physical-based model, or any other traditional approach. The evolution of ML models for both short and long-term flood prediction is illustrated in fig 4 [10].

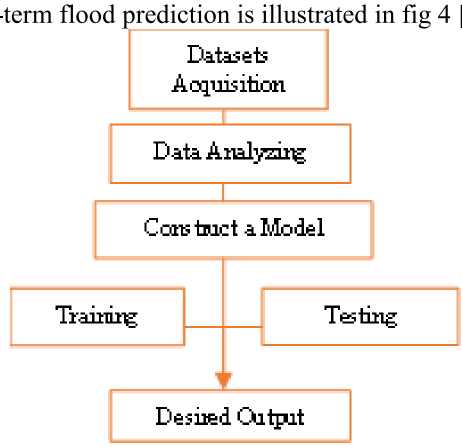

Fig 3 Basic flow for building the ML model [10]

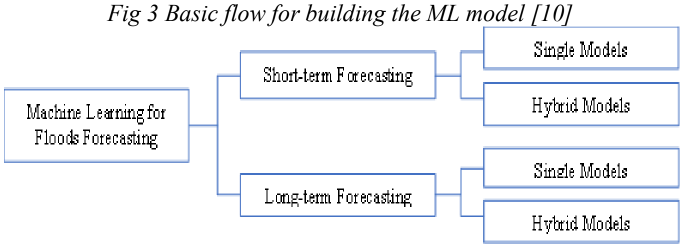

Fig 4 ML model developments [10]

B. Deep Learning (DL)

DL is a powerful category of neural networks that have evolved to have more hidden layers and a more complex architecture than their predecessors (i.e., Multilayer Perceptron). While applying DL, a supervised, semi-supervised, and unsupervised approach can be taken. DL may be broken into two critical subfields: feedforward and recurrent neural networks. Both subfields are dependent on data flows. Each branch has different iterations and contributes to forming a wide range of advanced networks such as ResNet and many others [11]. This study considers these advanced AI techniques for flood forecasting in tropical regions worldwide.

IV. ADVANCED AI FOR FLOOD FORECASTING

In this paper, advanced AI techniques are based on their application for flood forecasting. This section first briefly describes the working mechanism of these techniques and later a few studies from the literature for assessing these AI techniques.

C. Support Vector Machines(SVMs)

SVM was developed by [12] to get around problems associated with making predictions. The goal of SVM is to reduce the generalisation error, while ANN aims to minimise the training error. These methods achieve the highest possible degree of precision in their respective spheres. In contrast to ANN, SVM functions according to the principle of structural risk minimisation rather than empirical risk minimisation [13]. The primary working mechanism of the SVM is shown in fig 5.

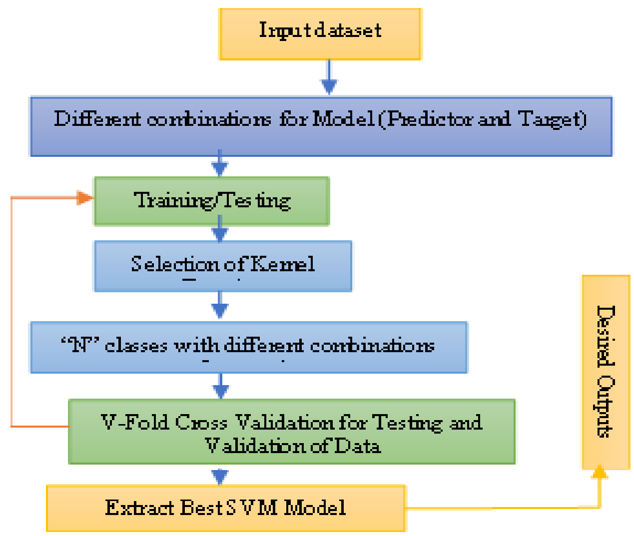

Fig 5 Schematic representation of SVM

D. ARTIFICIAL NEURAL NETWORKS (ANNS)

An ANN model aims to establish a connection between the input variables and the outcomes predicted for the future without using predefined coding. It is suitable for drought forecasting because of the random nature of drought characteristics and the systems' complexity [14]. This network has three layers: an input layer, one or more hidden layers, and an output layer shown in fig 6.

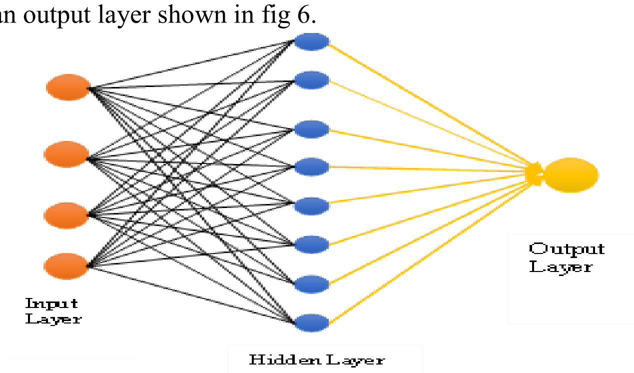

Figure 6 ANNs basic structure

A. Adaptive neuro-fuzzy inference system (ANFIS)

Instead of numerical uncertainty, the Fuzzy Logic (FL) technique is based on language uncertainty expression. A fuzzy inference system (FIS) has three components. (1) fuzzy if-then rules, (2) a membership Function database, and (3) an inference system that combines fuzzy rules and produces system outputs. Fuzzy logic has no way of finding model parameters and building fuzzy rules. ANFIS, a mixture of ANN and FL techniques, has shown promise to capture the advantages of both methods within a single organisational framework (fig 7). ANFIS overcomes the fundamental difficulty in fuzzy system design by employing ANN's learning power to develop fuzzy rules and optimise their parameters[15].

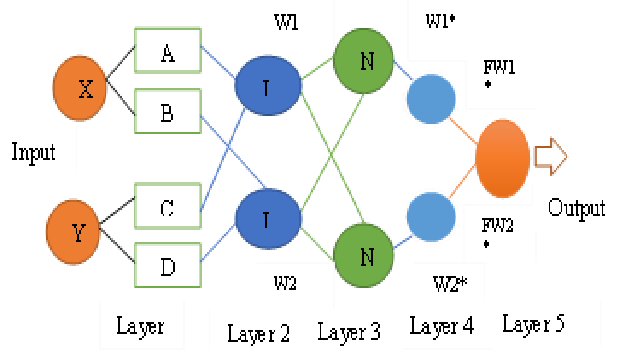

Figure 7 ANFIS basic structure

B. K-Nearest Neighbors (KNNs)

KNN is an algorithm for supervised machine learning that can address classification and regression issues. The algorithm is straightforward, yet it needs to be labelled training data. When data points are close together, or there is a noticeable difference between different classes, the KNN algorithm marks them as belonging to the same group (Fig 8). KNN uses the Euclidean distance formula to distinguish between two data points on a graph. One benefit of KNN for flood forecasting is that the data points can be easily trained and classified without much tuning [7].

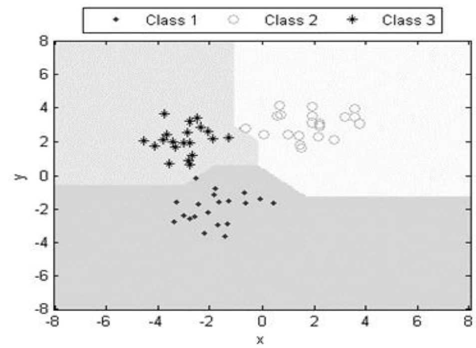

Figure 8 Simple KNN classifier [7]

Table 1 shows the previously employed AI techniques with different combinations of models described in the earlier section. These studies show that AI is the most precise and exact method for flood forecasting in other regions using different variables.

Table 1 A summary of the flood modelling AI techniques accessible in the scientific literature.

AI Technique	Variable	Forecast Type	Territory	Outcomes	Reference
Hybrid AI system (Modular ANN composed MLPs.)	1. Daily record of river flow volume (m^3/s) 2. Level of rainfall (ml/m2) 3. Temperature (◦C)	Short time prediction	Colombia	AI technologies can make predictions about univariate time series. Still, a correct neural network requires enough data and a daily update of the case base.	[3]
Hybrid Recurrent Neural Network	1. Outflow Flow 2. Rainfall	Flood trends and peak times.	China	Compared to more conventional hydrological models and machine learning networks, the ML model consistently predicts the peak time and overall flood volume with the least amount of error possible.	[14]
SVM, LSTM network and (RNNs)	Past Hydrological Data	Water level forecasting	Korea	All three water level forecasting algorithms performed well in the analysis. LSTM network performance rose as the earlier data correction period increased.	[5]
ANFIS Model and ANNs	Rainfall and Runoff Data	Short time prediction	India	The relative peak flow error of the fuzzy neural model was within a reasonable range. The neuro-fuzzy model could predict 47.95% of flow values 1 hour in advance with an error of less than 1% relative. In contrast, the neural network and the fuzzy model predicted 36.96%	[4]

				and 18.89%, respectively.	
Soil and Water Assessment Tool (SWAT) and ANNs	1. Annual Discharge 2. Base Flow 3. Q_{dry} 4. Q_{wet}	Daily streamflow	Nepal	SWAT model simulated low flow better than ANN in the three river basins. ANN model performed better for the high flows.	[6]
SVM and ANNs	1. Rainfall 2. Temperature	Hydrological Drought Forecasting	Australia	ANN predicted drought trends better than SVR, with an R2 of 0.86 against 0.75. Sea surface temperatures and the climatic indicator (Pacific Decadal Oscillation) had little effect on temporal drought.	[13]
k-nearest neighbour (KNN)	Hydrological historical data	Real-time Flood Forecasting	China	More exact forecasts were produced using area-weighted rainfall as an input to ANNs and MNLR and geographically distributed precipitation as an input to ANFIS and MLR.	[16]
ANN, ANFIS, and Regression Models	Rainfall	Short time prediction	Iran	Area-weighted rainfall as an input to ANNs and MNLR and spatially distributed precipitation to ANFIS and MLR led to more accurate forecasts.	[9]

IV. WARNINGS AND DISSEMINATION

The process of creating flood warnings and disseminating them has two primary steps: the first is the preparation of forecasts and alerts, followed by their transmission to end users, and the second is the response to them. The ability to supply quick and comprehensive coverage is essential to the success of any system. However, accurate forecasting without good communication and coordination will not result in the responses that the exposed society has planned. These responses include evacuating vulnerable communities (such as the aged or children who cannot quickly respond to warnings) and the mobility of assets (food, livestock, moveable goods, etc.). Forecasts and alerts can be disseminated via various communication channels, such as the internet, flood bulletins sent to emergency service agencies (such as the fire and police departments), television, radio, cellphone, bulk SMS, horns, flags, and social media. Other communication channels include flood newsletters sent to emergency service agencies (such as the fire and police departments).

V. CONCLUSION

Although several models have been developed and put into use for flood forecasting in various countries, there is still a need for computationally efficient models that can adequately consider catchment properties and the variations in those properties, and that have the capability of having their forecasts quickly updated. Techniques from AI can fill this void. We will be able to respond more rapidly to crisis scenarios and save a significant number of lives if we use AI in conjunction with appropriate warning and dissemination mechanisms.

VI. ACKNOWLEDGMENT

The authors would like to express their gratitude to The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi and the Center of Excellence on Energy Technology and Environment (CEE), Ministry of Higher Education, Science (MHESI), Research and Innovation and Department of Mathematics for their financial and technical support provided to perform this study.

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